

Contact Plan Optimization and Distributed State Estimation for Delay Tolerant Satellite Networks

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Abstract—Future deep space exploration, particularly the upcoming lunar missions, stands to benefit significantly from networks of small satellites, given their cost-effective communication and monitoring capabilities. However, challenges like frequent network partitions, high communication delays, and the demand for onboard orbit determination (OD) and time synchronization present obstacles for these satellite networks. One of the key problems in this context is how to schedule satellite communications and generate a contact plan. This paper presents an integrative solution that combines navigation and communications performance for contact plan optimization. Additionally, we propose a novel two-way inter-satellite link (ISL) scheme to acquire range and differential clock bias estimates that take into account rapid inter-satellite range shifts and signal propagation delays. Building on these, a distributed orbit and clock bias estimation algorithm is introduced, which emphasizes the use of batch processing to improve orbit observability. Performance metrics are validated through a simulation of a lunar navigation constellation, assessing OD error, time synchronization error, and inter-satellite communication delay. The combined insights and proposals presented, especially in terms of navigation and communications performance tradeoffs, offer promising avenues to enhance the operability and efficiency of small satellite networks.

1. INTRODUCTION

Networks of small satellites will play a key role in future deep space exploration, supporting orbit and surface missions thanks to their cost-effective communication and monitoring capabilities. One representative example is NASA's Lunar Communications Relay and Navigation Systems (LCRNS) [1], a network of satellites providing networking and navigation services in the lunar region.

These satellite networks, however, are subject to frequent partitions and high delays due to the limitation in the resources of small satellite platforms to establish multiple inter-satellite links (ISLs), long distances between satellites, and signal blockage by planetary bodies. At the same time, future satellite networks in deep space will be desired to perform onboard orbit determination (OD) and time synchronization, with the limited capacity of the Deep Space Network. To fully exploit the potential of small satellite networks, onboard OD and timing algorithms are desired to be scalable and flexible to dynamic network topology. Therefore, the implementation of ISLs within the satellite network needs to be designed considering the two objectives: 1) facilitating distributed onboard OD and time synchronization via inter-satellite ranging, and 2) enabling data exchange among satellites with low communication delay.

Regarding the first objective, Sipowa et al. proposed a distributed OD framework for satellite swarms that combines Covariance Intersection (CI) with Unscented Information Filter (UIF) [2]. By using CI, each agent only needs to estimate and store their own state estimates and covariances, which makes the framework scalable. However, their work targeted relative OD, and assumed small separations between satellites, restricting it from being used for satellite constellations which typically have larger satellite separations. They also did not model nor estimate onboard clock offsets and drifts in their work, which needs to be corrected for satellite constellation operations.

Regarding the second objective, Yan et al. proposed a contact plan optimization algorithm that accommodates the requirements of both OD and communication [3]. They formulated and solved a constraint optimization problem that sets inter-satellite communication delay as the optimization objective while enforcing the satellites to switch ISL pairs at every timeslot for improved geometrical diversity of the ISL directions. They also developed a heuristics algorithm based on simulated annealing to solve the problem. However, the performance of the optimized plan was evaluated only for the communication delay, and not for OD or clock synchronization errors.

The goal of this paper is to extend and fuse these two lines of work to efficiently coordinate long-range ISLs within a small satellite constellation to meet both navigation and communication requirements. First, we optimize the contact plan solving for the Pareto front of ISL geometrical diversity and communication delay by incorporating the result of a covariance analysis simulation directly in the objective function. Second, we propose a two-way ISL scheme to obtain the range and differential clock bias estimates, considering rapid changes in the inter-satellite range and propagation delays. Finally, we propose a distributed orbit and clock bias estimation algorithm, which processes the two-way range and differential clock bias measurements in batch. The proposed framework can estimate absolute orbits instead of relative orbits, thanks to the improved observability of the proposed batch processing scheme. Finally, integrating the three segments together, we demonstrate the performance of the optimized contact plan and the distributed state estimation algorithm by evaluating the OD error, time synchronization error, and inter-satellite communication delay in a simulated lunar constellation.

The paper is arranged as follows. Section 2 introduces the background with the related works, while Section 3 formulates the contact plan design problem as a constrained optimization problem. Section 4 proposes a contact plan design method based on simulated annealing. Section 5

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proposes a two-way ranging and timing scheme that could be used for links with large separation and fast relative motion. Section 6 provides the details of the proposed distributed state estimation algorithm. Performance evaluation is provided in Section 7. The paper concludes with Section 8.

2. RELATED WORKS

Consensus and Diffusion Kalman Filters

The distributed estimation problem has been investigated extensively for the past two decades. Earlier work by Spyer et al. [4] proposed a decentralized Kalman Filter, in which local Kalman filters communicate with all other nodes to estimate the state of fully connected networks. Later, Olfati-Saber et al. [5] proposed a distributed Kalman Filter, where each agent only communicates with its neighbor to reach a consensus on estimates with other nodes on the network. While these consensus-based methods have excellent mathematical properties, they impose significant timing constraints on the network, requiring the network to run the consensus step at much higher rates than the filter updates [6]. To overcome this challenge, Cattivelli, et al. [7] proposed a diffusion Kalman filter, in which each node performs a single-step update using a convex combination of neighbor node estimates. However, these distributed filtering approaches assume that the states to be estimated among the agents are the same. This assumption does not apply to our scenario, where the primary interest of each satellite is in estimating its own state.

Covariance Intersection (CI)

Covariance Intersection (CI) [8] is an algorithm to consistently fuse two measurements with unknown correlations. In CI, we consider a scenario where we have two consistent estimates $\hat{a}$ and $\hat{b}$ of quality of interest x . Here, by consistent we mean that the covariance of the errors are smaller than the covariance of the estimates, as follows.

$$E[\hat{x}\hat{x}^T] - E[(\hat{x} - \bar{x})(\hat{x} - \bar{x})^T] \geq 0 \quad (1)$$

where $\bar{x} \in \mathbb{R}^n$ is the true value. Let the corresponding covariances of $\hat{a}$ and $\hat{b}$ and their cross-correlation $P_{aa}, P_{bb}, P_{ab} \in \mathbb{R}^{n \times n}$, respectively. If P_{ab} is known, the new estimate could be obtained by

$$\hat{c} = W_a \hat{a} + W_b \hat{b} \quad (2)$$

$$P_{cc} = W_a P_{aa} W_a^T + W_a P_{ab} W_b^T + W_b P_{ba} W_a^T + W_b P_{bb} W_b^T \quad (3)$$

where W_a and W_b are weights chosen to minimize the trace of P_{cc} , which leads to the weighted least-squares solution

$$\hat{c} = (H^T (C^J)^{-1} H)^{-1} H^T (C^J)^{-1} \hat{c}^J \quad (4)$$

$$P_{cc} = (H^T (C^J) H)^{-1} \quad (5)$$

where

$$\hat{c}^J = \begin{bmatrix} \hat{a} \\ \hat{b} \end{bmatrix}, \quad C^J = \begin{bmatrix} P_{aa} & P_{ab} \\ P_{ab}^T & P_{bb} \end{bmatrix}, \quad H = [I_n \quad I_n] \quad (6)$$

However, when P_{ab} is not known, if we ignore them by setting $P_{ab} = 0$, we get an overconfident estimate that is not consistent. To address this problem, the CI algorithm uses a convex combination of the two estimates to obtain

an improved consistent estimate $\hat{c}$ with covariance P_{cc} , as follows

$$P_{cc}^* = (\omega P_{aa}^{-1} + (1 - \omega) P_{bb}^{-1})^{-1} \quad (7)$$

$$\hat{c}^* = P_{cc}^* (\omega P_{aa}^{-1} \hat{a} + (1 - \omega) P_{bb}^{-1} \hat{b}) \quad (8)$$

where ω is a parameter optimized to minimize the trace or determinant of P_{cc} . CI has been applied for decentralized multi-robot cooperative localization and sensor fusion problems [9, 10]. The overview of CI is shown in Figure 1.

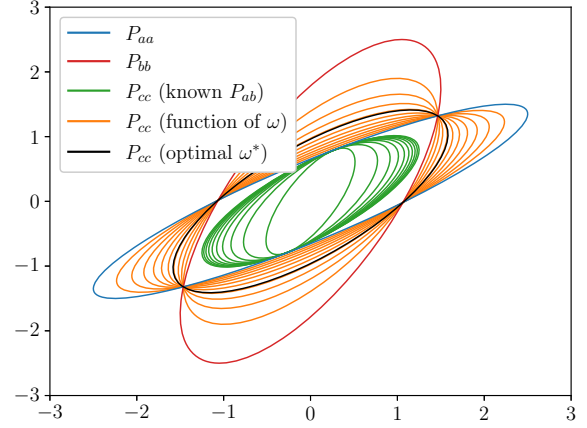


Figure 1: The overview of Covariance Intersection (CI), replicated from [2]. In CI, sensor estimates with covariance P_{aa} and P_{bb} with unknown correlation P_{ab} are combined to obtain the consistent and conservative covariance P_{cc} of the new estimate c . The weight ω to combine P_{aa} and P_{bb} is optimized to minimize the determinant of P_{cc} .

Distributed Spacecraft Orbit Determination

Various multi-agent frameworks have been proposed for collaborative orbit determination of networked spacecraft swarms with different sensor suites. Example frameworks include DiGITAL [11] system for nanosatellites with multi-GNSS receivers, Angles-only Absolute and Relative Trajectory Measurement System (ARTMS) [12] for angles-only navigation with optical sensors, and Autonomous Nanosatellite Swarming (ANS) [13] framework which simultaneously conducts spacecraft state estimation and characterization of the unknown gravity field of the orbiting asteroid. In these works, the states of all spacecraft within the swarm are stacked as a vector with a size of ρN , where ρ is the number of states and N is the number of satellites in the swarm. While this formulation enables accurate handling of the correlations between state estimates of each satellite in the swarm, they are not scalable to large swarm sizes, as the computational cost of the Kalman Filter grows square with respect to the state size.

To reduce the state size to be estimated in spacecraft swarm navigation, Sipowa et al. applied CI for spacecraft swarm navigation [2]. In their framework, CI was combined with the Unscented Information Filter (UIF). By using CI, each agent only needs to estimate and store their own state estimate and covariances, which makes the framework scalable. The algorithm was demonstrated to work robustly in a relative orbit determination scenario with 5 satellites, where each satellite takes the range, range rate, and orientation measurement with respect to the other satellite.

Orbit Determination with Inter-satellite Two-way Range and Time Difference Measurement

The work by [2] only estimated the relative state with respect to the reference satellite and assumed that the orbit of the reference satellite is known by all satellites. However, previous works [14, 15] showed that when there are a sufficient number of measurements and/or perturbations in the dynamics, the absolute orbit determination problem with only relative range or bearing measurement becomes observable [16].

One caveat of the above works, which simulates absolute orbit determination with inter-satellite range measurement, is that they do not model the two-way ranging procedure in detail and assume that instantaneous range, clock bias difference, (and range-rate) measurement could be obtained within certain errors without delay. In reality, errors arising from light time delays, clock drifts, and changes in the crosslink range during signal propagation cannot be ignored, especially in large separation scenarios.

Mo et al. [17] proposed a scalable multi-satellite scheme to obtain range and clock bias difference measurement. The authors employed time-division multiple access (TDMA) mechanisms and asymmetric double-sided two-way ranging (ADS-TWR). TDMA has advantages over FDMA and CDMA methods in terms of scalability, usage of space spectrum resources, and power and is well suited for small satellite platforms. However, ADS-TWR assumes small separation and relative motion, and therefore could generate large clock bias estimation errors when these effects become non-negligible.

Contact Plan Optimization

Different methods for designing contact plans for Global Navigation Satellite Systems (GNSS) have been proposed. Past research by Noakes et al. [18] centered on contact plans aimed at enhancing network connectivity through a process known as distributed optimization. Meanwhile, Chang et al. [18] applied a link assignment strategy aimed at minimizing the maximum link usage. The authors model the satellite network as a finite state automaton (FSA), where each state corresponds to an equal-length interval in the system period of the constellation, thus solving a set of link assignment problems in fixed topology networks. However, these studies [18, 19] were primarily concerned with the design of contact plans for communication satellite networks, which do not have to account for inter-satellite ranging demands or the temporal evolution of the network's topology.

Huang et al. [20] proposed methods for controlling the topology that could minimize the cost of the paths and maximize reliability in predictable DTNs. Similarly, Fraire et al. [21, 22] focused on creating fair and routing-aware contact plans, proposing several design algorithms. Nevertheless, these works mainly target communication satellite networks or remote sensing satellite networks and do not take into account the requirement for inter-satellite ranging.

Shi et al. [23] developed a contact plan that considered both crosslink ranging and data exchange requirements of ISLs in GNSS. However, the author of [23] did not have to account for the temporal evolution of the network's topology because each satellite could be equipped with three ISLs. Yan et al. [3] address the design of contact plans for navigation satellite networks with one ISL per satellite, taking into consideration both inter-satellite ranging and inter-satellite communication requirements. In recent work by Yan et al. [24], the authors

propose a multi-objective optimization model, using the position dilution of precision of links and the transmission time-delay as the navigation and communications metrics and use a genetic algorithm to optimize the contact plans for Earth GNSS constellations.

3. PROBLEM FORMULATION

The satellites will use ISL for both navigation and communication. In this section, we define the satellite states to be estimated, visibility between satellites based on geometric consideration, and the temporal topology handling scheme. Finally, we formulate the contact design problem that incorporates both navigation and communication objectives and constraints.

Satellite States

We consider a constellation of S spacecraft. We assume that each spacecraft is equipped with communication devices that permit information exchange between satellites, and a chip-scale atomic clock (CSAC). The state of satellite i consists of position r^i and velocity v^i in the body-centered inertial frame, and clock bias $\delta\tau^i$ and clock drift $\delta\dot{\tau}^i$, converted into meters by multiplying the light speed c .

$$x^i = [r^i \ v^i \ c\delta\tau^i \ c\delta\dot{\tau}^i] \quad (9)$$

Each spacecraft follows the perturbed two-body dynamics around the central body, which is represented as

$$\dot{v}^i = -\mu \frac{r^i}{|r^i|^3} + f_p(r^i, v^i) \quad (10)$$

where f_p is the perturbation forces from the J_2 term and third-body gravity from the Sun and the Earth. The clock bias and drift are modeled as a linear system with noises as shown below.

$$\begin{bmatrix} c\delta\tau^i \\ c\delta\dot{\tau}^i \end{bmatrix}_{k+1} = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} c\delta\tau^i \\ c\delta\dot{\tau}^i \end{bmatrix}_k + w_{clk} \quad (11)$$

where the process noise covariance is given by [25]

$$Q_{clk} = c^2 \begin{bmatrix} \sigma_1^2 \Delta t + \sigma_2^2 \frac{\Delta t^3}{3} & \sigma_2^2 \frac{\Delta t^2}{2} \\ \sigma_2^2 \frac{\Delta t^2}{2} & \sigma_2^2 \Delta t \end{bmatrix} \quad (12)$$

where σ_1, σ_2 is the Allan variance power spectrum of the white noise and flicker noise. We used $\sigma_1 = 3.0 \times 10^{-10}$, $\sigma_2 = 3.0 \times 10^{-12}$ retrieved from the Allan variance of MICROSEMI CSAC.

We use TDMA to coordinate the communication process between satellites. Each transmitter-receiver pair will be assigned a time slot to communicate with the other satellite based on the contact plan. The detailed mathematical formulation of the measurement equations is derived in Section 6.

We have two types of satellites: the anchor satellites (nodes) and the non-anchor satellites. The anchor nodes have access to an external navigation source (e.g. GPS, ground station links) that can be used to continuously update its orbit and clock bias estimates. Other satellites (e.g. non-anchor nodes) rely on ISL range and relative timing measurement to estimate their orbit and clock bias. The goal of each agent is to obtain a sequential state estimate for their own state x^i , without estimating the states of other spacecraft in the swarm x^j . This is achieved by the combination of an EKF and CI, as it will be explained in Section 6.

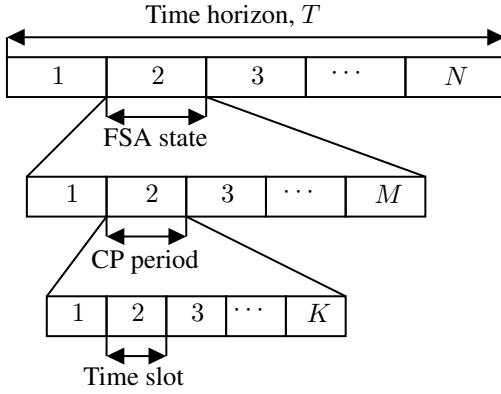


Figure 2: Topology handling scheme, including finite state automaton (FSA) states, contact plan (CP) periods, and time slots.

Topology Handling Based on FSA

The constellation network topology changes periodically due to the movement of the satellites and can be modeled as an FSA [26]. The *FSA states* are derived by dividing the system period of the satellites T into N equal-length intervals, disregarding those satellites that are only partially visible within a given state. In other words, two satellites are said to be visible from each other in a state if they are within line-of-sight throughout the entire state duration. Therefore, the contact plan design can be performed in each FSA state based on fixed topology.

Satellites require frequent communication to increase the precision of autonomous navigation and typically have fewer ISLs than the number of visible satellites. Therefore, they need to switch their ISLs to enable inter-satellite ranging with different satellites. As in previous work [24], we further divide one FSA state into M equal-length time intervals denoted as *contact plan periods*.

Every contact plan period is subsequently split into K time slots of the same length. The way the topology is managed is illustrated in Figure 2. Depending on the time slot, navigation satellites establish ISLs with different visible satellites. The contact plan design is carried out within each FSA state, and all contact plan periods within a single FSA state adhere to the same contact plan. In other words, contact plans are created considering the fixed topology of an FSA state and determine the ISLs to be established during each time slot, repeating the plan for every contact plan period in the FSA state.

State Estimation Accuracy Calculation Based on EKF Covariance Analysis

The states of each satellite are estimated using the distributed state estimation algorithm with batch CI, which will be explained in detail in Section 6. When evaluating the navigation performance of a given contact plan during contact plan optimization, to save computational cost, we run a covariance analysis with inflated noise instead of actual distributed state estimation with CI. The procedure of the covariance analysis is detailed below.

We define timestep k as the beginning of time slot $k \in [1, KM]$. Every timestep, each satellite performs a time

update as follows

$$P_{k|k-1}^i = \Phi_{k|k-1}^i P_{k-1|k-1}^i \Phi_{k|k-1}^{i\top} + Q_k^i \quad (13)$$

$$\Phi_{k|k-1}^i = \begin{bmatrix} \Phi_{k|k-1}^{i,rv} & 0_{6 \times 1} & 0_{6 \times 1} \\ 0_{1 \times 6} & 1 & \Delta t \\ 0_{1 \times 6} & 0 & 1 \end{bmatrix} \quad (14)$$

$$Q_k^i = \begin{bmatrix} Q^{rv} & 0_{6 \times 2} \\ 0_{2 \times 6} & Q^{clk} \end{bmatrix} \quad (15)$$

where P^i is the covariance estimate of satellite i , $\Phi_{k|k-1}^i$ is the state transition matrix (STM) of satellite i state from timestep $k-1$ to k , and Q^{rv} is the process noise matrix for satellite position and velocity. Note that since we are not updating the state estimates in covariance analysis, the STM $\Phi_{k|k-1}^{i,rv}$ can be precomputed and stored before the optimization using the true trajectory.

When satellite i and satellite j perform ISL at time slot k , we perform the measurement update at both ends. The ISL measurement $y_k^{i,j}$ consists of inter-satellite range and differential clock bias. The details of how we obtain these measurements are explained in Section 5, but in the optimization, these are modeled as instantaneous range and differential clock bias measurements at the beginning of each time slot. The measurement, Jacobian, and the noise matrix are set as follows

$$y_k^{i,j} = \begin{bmatrix} \|r_k^i - r_k^j\| \\ c(\delta t_k^i - \delta t_k^j) \end{bmatrix} \quad (16)$$

$$H_k^{i,j} = \begin{bmatrix} -\left(\frac{r_j - r_i}{\|r_j - r_i\|}\right)^\top & 0_{1 \times 3} & 0 & 0 \\ 0_{1 \times 3} & 0_{1 \times 3} & -1 & 0 \end{bmatrix} \in \mathbb{R}^{2 \times 8} \quad (17)$$

$$R_k^{i,j} = \begin{bmatrix} \sigma_r & 0 \\ 0 & \sigma_t \end{bmatrix} + (-H_k^{i,j}) P_{k|k}^j (-H_k^{i,j})^\top \in \mathbb{R}^{2 \times 2} \quad (18)$$

The second term in (18) is the additional noise from the state knowledge uncertainty of the ISL pair satellite j . Note that the cross-covariances between satellites i and j estimates are not considered in (18), so even with this additional noise, the obtained covariance is still optimistic. However, it could still be used as a metric to compare different contact plan topologies.

The covariance update equation for the measurement update at satellite i is written as

$$K_k^i = \alpha P_{k|k-1}^i (H_k^{i,j})^\top \left[R_k^{i,j} + H_k^{i,j} P_{k|k}^j (H_k^{i,j})^\top \right]^{-1} \quad (19)$$

$$J_k^i = I_{8 \times 8} - K_k^i H_k^{i,j} \quad (20)$$

$$P_{k|k}^i = J_k^i P_{k|k-1}^i (J_k^i)^\top + K_k^i R_k^{i,j} (K_k^i)^\top \quad (21)$$

where α is the weight to adjust the Kalman gain. We use $\alpha = 1/2$ when the ISL pair is not an anchor node to avoid double counting on the same measurements. $\alpha = 1$ is used when the satellite j is an anchor node, which will not update its state using ISL measurements.

Starting from a predefined initial covariance $P_{0|0}$, We repeat these updates for the entire FSA state, $k = 1, \dots, K \times M$. At

each time step, for each satellite, we compute the square root of the trace of the covariance of the position and clock bias as follows to evaluate the positioning and timing performance.

$$P_{trace}(i, k) = \sum_{l \in \{1, 2, 3, 7\}} P_{k|k}^i(l, l) \quad (22)$$

where $P_{k|k}^i(l, l)$ shows the l th diagonal element of the matrix $P_{k|k}^i$.

Formulation of the Contact Plan Design Problem

In this section, we formulate the contact plan design problem as a constrained optimization problem. The contact plan for the navigation satellite network needs to meet the needs of both inter-satellite ranging and inter-satellite communication, which translates into a desired high number of ISL switches and low communication delay. To account for both objectives, we set the objective as a weighted sum of the inter-satellite communication delay and the simulated navigation error from the covariance analysis, and formulated the contact plan design problem as follows:

$$\text{minimize } F = w_{comm}F_{comm} + w_{nav}F_{nav} \quad (23)$$

$$\text{subject to} \quad (24)$$

$$u_{i,j,k} = u_{j,i,k}, \quad \forall i, j, k \quad (25)$$

$$v_{i,j} - u_{i,j,k} \geq 0, \quad \forall i, j, k \quad (26)$$

$$\sum_{j=1}^S u_{i,j,k} = 1, \quad \forall i, k \quad (27)$$

$$\sum_{k=1}^K u_{i,j,k} \leq N_{max}, \quad \forall i, j \quad (28)$$

$$u_{i,j,k} \in \{0, 1\}, \quad \forall i, j, k \quad (29)$$

$$i, j = 1, 2, \dots, S, \quad k = 1, 2, \dots, K$$

where

$$F_{comm} = \frac{1}{K} \frac{1}{S(S-1)} \sum_{k=1}^K \sum_{i=1}^S \sum_{j=1, j \neq i}^S d(i, j, k) \quad (30)$$

is the average network delay and accounts for communication performance,

$$F_{nav} = \frac{1}{K} \frac{1}{S} \sum_{i=1}^S \sum_{k=1}^K \sqrt{P_{trace}(i, k)} \quad (31)$$

is accounts for the navigation performance, S is the number of satellites, K is the number of time slots, $v_{i,j}$ is the visibility between satellite i and j within a certain FSA state ($V = [v_{i,j}]$ is the visibility matrix), $u_{i,j,k}$ is the ISL connectivity between satellite i and j within time slot k ($U = [u_{i,j,k}]$ is the connectivity matrix, which is also the contact plan and decision variable), $d(i, j, k)$ is the cost of the communication—in this case, the best delivery time (BDT) between satellite i and j starting at time slot k , P_{kf} is the covariance based on covariance analysis, and w_{comm} and w_{nav} are the weights of each objective.

If satellites i and j are visible during the entire duration of an FSA state, $v_{i,j} = 1$, and they will establish an ISL at the time slots k such that $u_{i,j,k} = 1$. Constraint (25) enforces the communication between two satellites to be

bidirectional, (26) ensures that the ISL is only established if the two satellites are visible to each other, (27) guarantees that each satellite establishes one and only one ISL within each time slot, and (28) limits the maximum number of ISLs that can be established between two satellites during a contact plan period to N_{max} in order to enforce ISL switches.

The BDT between two satellites $d(i, j, k)$ can be calculated using a Contact Graph Dijkstra Search (Algorithm 2, Appendix 8), a modified Dijkstra search that finds the best route through a contact graph [27]. The main advantage of expressing the contact plan as a contact graph data structure is that it can be used as input for traditional shortest-path algorithms such as the one presented in this work.

The core difference compared to the previous work [24] is the navigation metric F_{nav} . Compared to calculating PDOP at each timestep, the proposed metric, which uses covariance analysis, have multiple benefits. First, it can be applied when we have less than 3 ISLs in each time slot. Second, it can better capture the relationship between dynamics (time update) and the measurement that occurs when we batch process or filter measurements over time. Finally, it can value having more connections with satellites that have better state knowledge.

4. CONTACT PLAN OPTIMIZATION BASED ON SIMULATED ANNEALING

Considering that the contact plan problem defined in the previous section would rapidly become computationally intractable, we use a heuristic algorithm based on simulated annealing, a stochastic optimization method that mimics the slow cooling process in metallurgy to gradually reduce the search space and decrease the likelihood of accepting sub-optimal solutions. This approach has been demonstrated to be effective in topology design problems [19, 22], enabling the acquisition of sub-optimal solutions. Algorithm 1 presents the algorithm for the contact plan design based on simulated annealing. We begin by creating a random initial contact plan from which we will derive a new contact plan through branch and exchange methods [19, 22]. The branch and exchange method consists of exchanging two links while maintaining constraint satisfaction.

Then, we calculate and compare the objective functions of the new and the current contact plans. If the objective function of the new contact plan is improved, we approve the new contact plan and reduce the temperature T . Otherwise, we accept the new contact plan with a probability $P = \exp(-(F_{new} - F_{curr})/T)$, where F_{new} and F_{curr} are the objective functions of the new and the current contact plans, respectively. If the new contact plan is accepted, we reduce the temperature T by a cooling rate r . The algorithm stops when a set number of iterations N_{it} is reached.

5. TWO-WAY RANGING WITH JOINT DOPPLER MEASUREMENT

In this section, we derive the two-way range and clock bias measurement for inter-satellite links with large separation and relative velocity. The overview of the two-way crosslink procedure is illustrated in Figure 3. In the example, the transmitter satellite A sends the signal at time t_1 . After the receiver satellite B receives the signal at time t_2 , it retransmits to satellite A at time t_3 with delay (we set this delay

Algorithm 1 Contact plan design based on simulated annealing.

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1: procedure CONTACTPLANDSIGN( $V, T, r, N_{it}$ )
2:    $U \leftarrow \text{InitialContactPlan}(V)$ 
3:    $F_{best} \leftarrow \text{ObjectiveFunction}(U)$ 
4:   for  $i \leftarrow 1$  to  $N_{it}$  do
5:      $U' \leftarrow \text{GetNeighbor}(U)$ 
6:      $F' \leftarrow \text{ObjectiveFunction}(U')$ 
7:     if  $F' \leq F_{best}$  then
8:        $U_{best} \leftarrow U \leftarrow U'$ 
9:        $F_{best} \leftarrow F'$ 
10:    else if  $\exp((F_{curr} - F')/T) \leq \text{rand}(0, 1)$  then
11:       $U \leftarrow U'$ 
12:       $T \leftarrow r \cdot T$ 
13:    else
14:       $U \leftarrow U_{best}$ 
15:    end if
16:  end for
17: end procedure

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to 0.5 seconds in our simulation), which is received at satellite A at time t_4 . Satellite A will send back the time stamps to satellite B, where range and clock bias difference estimates will be calculated.

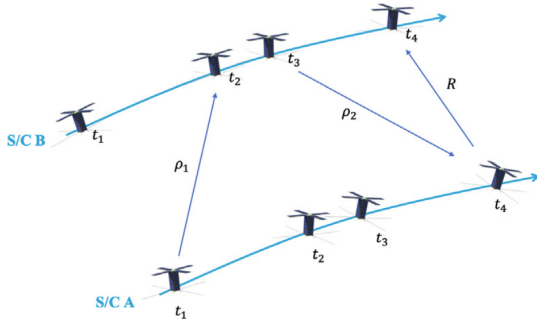


Figure 3: Two-way inter-satellite ranging scheme between two spacecraft (S/C), retrieved from [16].

Let $T_{A1}, T_{B2}, T_{B3}, T_{A4}$ the local clock time of satellite A and B at global time t_1, t_2, t_3 , and t_4 , respectively, which represents the true time the signal is receiver or transmitted. The local clock time is related to true time as follows

$$T_{A1} = t_1 + \delta\tau_{A1} \quad (32)$$

$$T_{B2} = t_2 + \delta\tau_{B2} + w_{r2} \quad (33)$$

$$T_{B3} = t_3 + \delta\tau_{B3} \quad (34)$$

$$T_{A4} = t_4 + \delta\tau_{A4} + w_{r4} \quad (35)$$

where $\delta\tau$ represents the clock bias and w_r is the time delay in the receiver due to thermal noise. We set the thermal noise error to 1.0 m (when converted to range via light speed). Combining the observed time stamps listed above, the goal of the two-way crosslink is to obtain the range and clock bias difference between satellites A and B at the following reference epoch

$$t_{ref} = \frac{T_{A1} + T_{A4}}{2} \quad (36)$$

Obtaining the reference range estimate from time stamps is straightforward and is commonly done in the literature. The

two range estimates from the observed timestamps are given by

$$\rho_{12} = c(T_{B2} - T_{A1}) \quad (37)$$

$$\rho_{34} = c(T_{A4} - T_{B3}) \quad (38)$$

From first-order Taylor expansion, the range between satellite A and satellite B at the reference time is given by [28]

$$\rho_{ref}^{AB} = \frac{1}{2}(\hat{\rho}_{12} + \hat{\rho}_{34}) \quad (39)$$

Obtaining the clock bias differences from the timestamps is more involved because we have to consider the errors coming from light time delay, clock drift, and the range rate. Previously proposed methods such as ADS-TWR assumed close separation between satellites and ignored these effects, which become non-negligible in large separation formations [17]. To cancel these errors, we propose a new formulation that compensates for the intersatellite rate changes, using the range rate estimate obtained from the frequency shift measurement.

We start with the following two equations

$$\begin{aligned} \rho_{12} &= \|r_A(T_{A1}) - r_B(T_{B2})\|_2 \\ &= c(t_2 - t_1) \\ &= c(T_2 - \delta\tau_{B2} - T_1 + \delta\tau_{A1}) + w_{12} \\ &\approx \rho_{ref} - \dot{\rho}_{ref}(t_{ref} - t_{A1}) \end{aligned} \quad (40)$$

$$\begin{aligned} \rho_{34} &= \|r_A(T_{A4}) - r_B(T_{B3})\|_2 \\ &= c(t_4 - t_3) \\ &= c(T_4 - \delta\tau_{A4} - T_3 + \delta\tau_{B3}) + w_{34} \\ &\approx \rho_{ref} + \dot{\rho}_{ref}(t_{A4} - t_{ref}) \end{aligned} \quad (41)$$

Where ρ_{12}, ρ_{34} is the true range, and w_{12}, w_{34} are range estimate errors coming from receiver thermal noise and clock drift, and $\dot{\rho}_{ref}$ is the inter-satellite range rate, evaluated at t_{ref} . By observing the Doppler shift $\omega_{rt} = \omega_r/\omega_t$ of the received signal at satellite A, we can calculate the range rate of the signal as follows.

$$\dot{\rho}_{ref} \approx \frac{c\omega_{rt}}{2 + \omega_{rt}} \quad (42)$$

For this paper, we assumed that we could get the range rate information with an error of 1 mm/s (1σ). Using this observed range rate, and plugging into Eq.(40) and Eq.(41), we obtain two clock bias difference estimates

$$\begin{aligned} \Delta\delta\tau_{12}^{AB} &= (\delta\tau_{ref}^A - \delta\tau_{ref}^B)_{12} \\ &= -(T_{B2} - T_{A1}) + \frac{\rho_{ref} - \rho_{ref}(t_{ref} - T_{A1})}{c} \end{aligned} \quad (43)$$

$$\begin{aligned} \Delta\delta\tau_{34}^{AB} &= (\delta\tau_{ref}^A - \delta\tau_{ref}^B)_{34} \\ &= (T_{A4} - T_{B3}) - \frac{\rho_{ref} + \rho_{ref}(T_{A4} - t_{ref})}{c} \end{aligned} \quad (44)$$

The clock bias difference estimate is calculated as the average of these two values

$$\Delta\delta\tau_{ref}^{AB} = \frac{1}{2}(\Delta\delta\tau_{12}^{AB} + \Delta\delta\tau_{34}^{AB}) \quad (45)$$

The advantage of this method is that by averaging the two measurements, we can cancel out the majority of the errors

coming from clock drift and range rates. In summary, the two measurements obtained per each two-way link is

$$y_k^{i,j,obs} = [\rho_{ref}^{AB} \quad \Delta\delta\tau_{ref}^{AB}]^T \quad (46)$$

The obtained measurement error is summarized in Figure 4. The obtained range and clock bias error are on the same scale as the thermal noise of the receiver, with errors from range-rates and clock drifts being mostly removed.

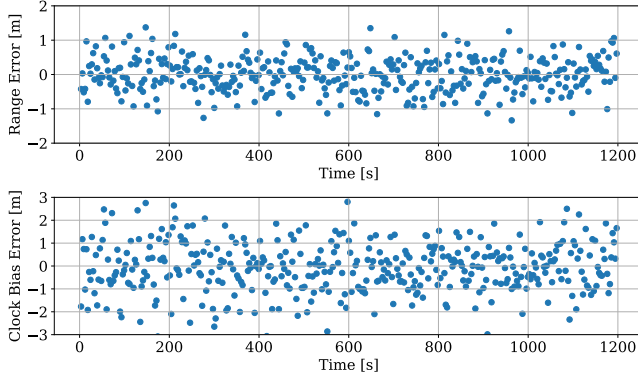


Figure 4: Simulated range and clock bias measurement error with the proposed two-way range with joint Doppler two-way scheme

6. DISTRIBUTED STATE ESTIMATION

Each satellite performs ISL following the optimized plan. All satellites except the anchor nodes process the two-way range and the differential clock bias measurement obtained by ISL in a distributed manner. Since the satellite does not estimate or store the cross-covariance estimates with respect to other satellites in the constellation, applying a simple Kalman Filter update that ignores the cross-terms can result in inconsistent estimates. Therefore, we use CI in the measurement update to obtain a conservative covariance estimate.

Each satellite estimates its state at the beginning of each time slot t_k . Note that estimates come with a delay since the observable is unavailable until the reception of the two-way signal. In the measurement update scheme using CI, we replace the measurement update equation (19), (20), (21) with the following equations [29]

$$\tilde{H}_k^i = (H_k^i)^\top \Phi^i(t_{ref,k}, t_k) \quad (47)$$

$$S_k^i = \frac{1}{\omega} \tilde{H}_k^i P_{k|k-1}^i (\tilde{H}_k^i)^\top + \frac{1}{1-\omega} R_k^i \quad (48)$$

$$K_k^i = \frac{1}{\omega} P_{k|k-1}^i (\tilde{H}_k^i)^\top (S_k^i)^{-1} \quad (49)$$

$$P_{k|k}^i = \frac{1}{\omega} P_{k|k-1}^i - K_k^i S_k^i (K_k^i)^\top \quad (50)$$

$$x_{k|k}^i = x_{k|k-1}^i + K_k^i (y_k^{i,obs} - y_k^{i,pred}) \quad (51)$$

Above, $t_k < t_{ref,k} < t_{k+1}$ is the reference time (36) of the measurement obtained at time slot k , and $\Phi^i(t_{ref,k}, t_k)$ is the state transition matrix of satellite i from the beginning of slot k to $t_{ref,k}$. $y_k^{i,pred}$ is the predicted measurement from the current state estimate. Other notations are the same of (19), (20), (21), and we dropped the superscript

j to simplify the notation. $\omega \in [0, 1]$ is a variable that is optimized to minimize the determinant of the posterior covariance estimate, as follows.

$$\omega = \text{minimize}_{\omega} P_{k|k}^i(\omega') \quad (52)$$

As seen in equation (48), ω is determined to balance the two covariances: the apriori state estimate covariance mapped into the measurement space (first term), and the inflated measurement covariance (second term). The variable ω takes a value closer to 1 as the second term becomes larger than the first term, resulting in a smaller update.

In our setting, each satellite gets only the range measurement and differential clock bias with respect to the ISL pair. Therefore, the satellite cannot solve for its 8 states by just using a single ISL measurement. Thus, the information obtained from each measurement is low compared to the apriori estimate from the time update, and if each satellite processes the measurements one by one at each timestep, they end up getting ω values close to 1, which means the measurements will be discarded. To avoid this, we stack N_{batch} measurements over time as follows.

$$y_{k,N_b}^i = [(y_k^i)^\top \quad (y_{k+1}^i)^\top \quad \dots \quad (y_{k+N_b-1}^i)^\top]^\top \quad (53)$$

$$H_{k,N_b}^i = \begin{bmatrix} (H_k^i)^\top \Phi^i(t_{ref,k}, t_k) \\ (H_{k+1}^i)^\top \Phi^i(t_{ref,k+1}, t_k) \\ \vdots \\ (H_{k+N_b-1}^i)^\top \Phi^i(t_{ref,k+N_b-1}, t_k) \end{bmatrix} \quad (54)$$

$$R_{k,N_b}^i = \text{diag}(R_k^i, R_{k+1}^i, \dots, R_{k+N_b-1}^i) \quad (55)$$

The size of the batch N_b is decided based on the trade-off between the observability and the delay of estimate we get. Since the orbit and clock estimation is decoupled when we use two-way range measurements, we ran a separate filter for the orbit and clock estimation. We set $N_b = 5$ for the orbital states and $N_b = 3$ for the clock states.

Note that the batch scheme ignores the correlations between the measurement covariances $R_k^i, \dots, R_{k+N_b-1}^i$ (They are correlated because each R_k^i is inflated based on the state covariance estimate of the ISL pair, which can be correlated each other). However, we observed that the contribution from these correlation terms is negligible compared to the inflated measurement noises, and consistent estimates can be generated. We plan to analyze the effect of the correlations within the batch observables in more detail in our future work.

Using this batch of measurements, the state is updated at timestep k as follows

$$S_{k,N_b}^i = \frac{1}{\omega} H_{k,N_b}^i P_{k|k-1}^i (H_{k,N_b}^i)^\top + \frac{1}{1-\omega} R_{k,N_b}^i \quad (56)$$

$$K_{k,N_b}^i = \frac{1}{\omega} P_{k|k-1}^i (H_{k,N_b}^i)^\top (S_{k,N_b}^i)^{-1} \quad (57)$$

$$P_{k|k}^i = \frac{1}{\omega} P_{k|k-1}^i - K_{k,N_b}^i S_{k,N_b}^i (K_{k,N_b}^i)^\top \quad (58)$$

$$x_{k|k}^i = x_{k|k-1}^i + K_{k,N_b}^i (y_{k,N_b}^{i,obs} - y_{k,N_b}^{i,pred}) \quad (59)$$

The latest estimate at timestep $k + N_b - 1$ is obtained by performing the time update from timestep k .

7. SIMULATION RESULT

In this section, we first present the experimental setup, including the parameters of the satellite constellation and the topology handling scheme. Then, we present the results of the contact plan design based on simulated annealing, showing the tradeoffs in optimizing the contact plan. Finally, we verify the navigation performance of the optimized contact plans using a distributed state estimation algorithm.

Experimental Setup

We assess the performance of the contact plan design algorithm based on simulated annealing in the lunar constellation shown in Figure 5. The orbital elements of the three orbital planes are based on [30]. The constellation comprises a total of 12 satellites among three planes, with an actual anomaly offset between satellites of 90° . Each navigation satellite is only equipped with a single ISL, which is only subject to moon occultation communication constraints. The parameters of topology handling are shown in Table 2. There are a total of 72 FSA states in one system period, each divided into 25 contact plan periods. A contact plan period contains 8 time slots, each lasting 3 seconds. The orbital elements of each satellite are given in Table 1. The 3 anchor satellite nodes, which we assume are equipped with better navigation capabilities, have an initial position and clock bias error of 1 m, and a velocity and clock drift error of 1 mm/s. They also get direct measurements of their position, velocity, and clock bias, with errors of 1 m, 1 mm/s, and 1m (when converted to meters), respectively. The remaining 9 satellites have an initial position and clock bias error of 50 m, and a velocity and clock drift error of 50 mm/s.

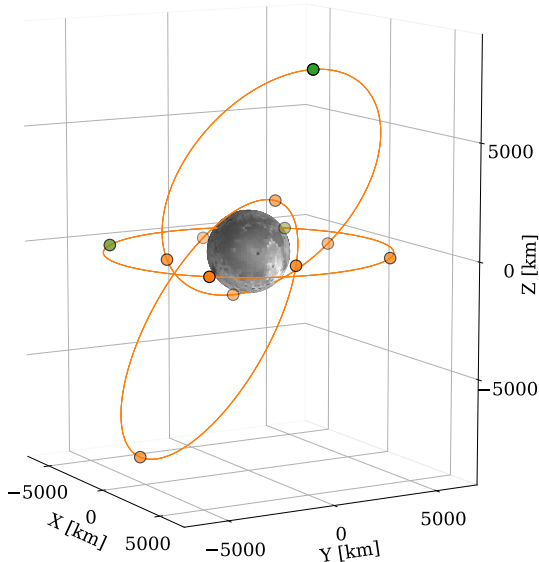


Figure 5: Constellation of 12 satellites around the Moon used for simulation. The points show the initial position of the satellites. The green points show the anchor nodes that have access to external positioning and navigation aids.

Figure 6 illustrates the number of visible satellites for the first satellite in each orbit. We observe that all three satellites have between 5 and 11 visible satellites in every FSA state.

Regarding the optimization algorithm based on simulated

Table 1: Orbital elements of the lunar constellation.

Orbital element	Plane 1	Plane 2	Plane 3
Semi-major axis a [km]	6142.4	6142.4	6142.4
Eccentricity e	0.001	0.6	0.6
Inclination i [deg]	0.001	57.7	57.7
Lon. of Asc. Node Ω [deg]	0	-90	0
Arg. of periapsis ω [deg]	-45	-90	90
True anomaly ν [deg]	(0, 90, 180, 270)		

Table 2: Parameters of topology handling based on FSA [3].

Parameter	Value
System Period (T)	12 hours
Number of FSA States (N)	72
FSA state duration	10 min
Number of contact plan periods (M)	25
Contact plan period duration	24 s
Number of time slots (K)	8
Time slot duration	3 s
Max same links allowed in K slots (N_{max})	5

annealing, we generate the initial random contact plan by solving the constraint satisfaction problem resulting from neglecting the objective function. Without the objective function, which can be computationally expensive and involves computing the shortest path in contact graphs as well as the navigation performance, the new problem is readily solved. In this case, we generate the initial plan by solving the constraint satisfaction problem using CVXPY [31].

We implemented the contact plan design based on simulated annealing in Algorithm 1 and the branch and exchange algorithm in Algorithm 3 in Python. The initial temperature is set to $T = 200$, the cooling rate is set to $r = 0.95$, and the number of iterations is set to $N_{it} = 1000$. We set an early stop criteria of 100 iterations without improvement in the objective function. It is important to mention that when computing the BDT, we consider the delay from the average distance between satellite pairs.

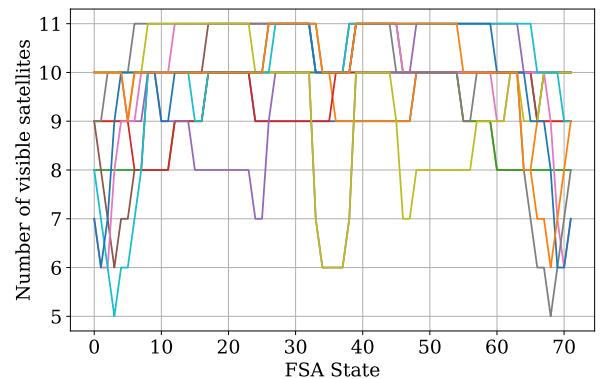


Figure 6: Number of visible satellites in all FSA states. Each color corresponds to a satellite in the constellation.

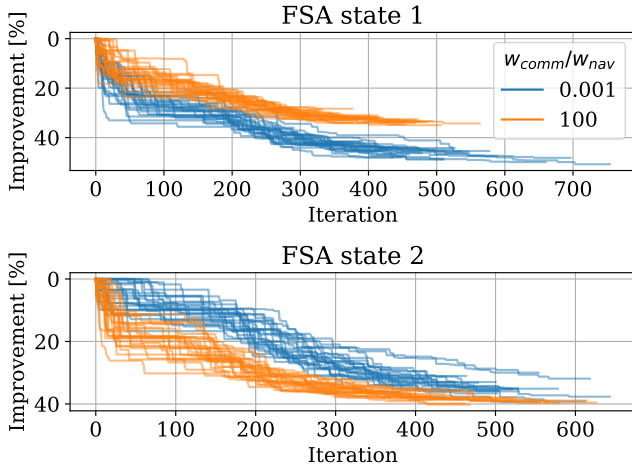


Figure 7: Objective value improvement vs. iteration number for two different FSA states and two different relative objective function weights.

Contact Plan Optimization

The proposed method is able to design a contact plan for the simulation use case successfully. Thanks to the initialization method based on constraint satisfaction and the implemented branch and exchange algorithm, we ensure that the contact plan is always feasible. Fig 7 shows the decrease of the objective function value throughout the optimization of two FSA states and two different relative objective weights. The algorithm tends to converge after approximately 500 iterations, as better plans are not found.

Figure 8a shows the scheduled ISLs for all the satellites of the constellation in the first time slot, and Figure 8b shows the scheduled ISLs for satellite 4 during a complete contact plan period of 8 time slots, obtained from the optimized contact plan. Fig 9 shows the evolution of the satellites reachable from satellite 4 over the first contact plan period, where we observe that all satellites are reachable after 5 time slots. The orange line shows the path that a message from satellite 4 would take to reach satellite 5 with minimum delay: it would have to go through satellites 3 and 9. This path involving four different satellites shows the complexity behind the objective function of the optimization problem.

Figure 10 shows the multi-objective optimization results for two FSA states. The results are obtained by using different weights in the objective function, w_{comm} and w_{nav} , as well as different random initializations of the contact plan. We observe how our method is able to explore the solution space and trace out the Pareto front. We highlight the tradeoff between communication and navigation performance, where the communication performance is measured by the average delay in the network in the form of BDT, and the navigation performance is measured by the position accuracy achieved with the scheduled crosslinks.

Contact plans optimized for communications performance present a BDT of 4.5 seconds and a positioning error of over 10 meters. On the other hand, contact plans optimized for navigation performance present a BDT of 6 seconds (25% worse) and a positioning error of 6 meters (40% better).

This difference is due to the link diversity in the contact plan. We observe that in contact plans optimized for navigation

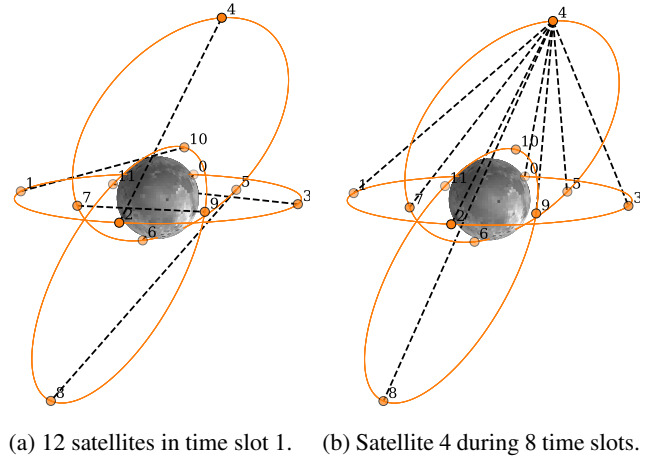


Figure 8: ISLs between satellites in the selected constellation are obtained from the optimized contact plan.

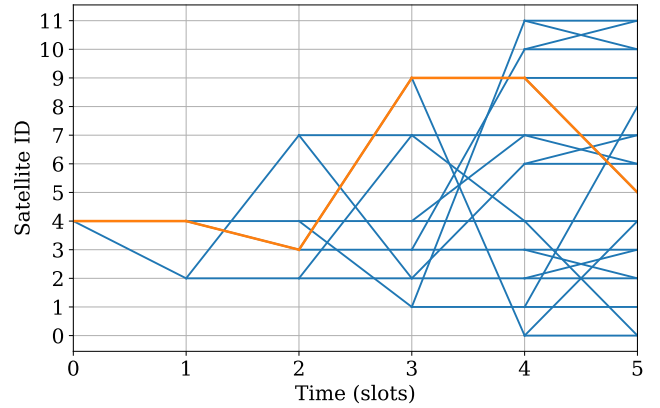


Figure 9: Possible routes starting at satellite 4. All the satellites are reachable after the first five time slots. In orange, we show the fastest route between satellite 4 and satellite 5, which includes a total of six different satellites.

performance, the number of unique links, i.e., with different satellites, in a contact plan period is 20% higher than in contact plans optimized for communications performance. This is in agreement with the hypothesis that, for inter-satellite ranging, satellites should frequently change their ISL targets, thus securing more diversity in the ISL direction and enabling superior precision in autonomous navigation. In contrast, inter-satellite communication necessitates the navigation satellite to establish a more stable link topology to efficiently transfer the packets for a given graph structure. Therefore, these two contrasting requirements could guide the design of the contact plan in divergent directions, and it is necessary to consider both objectives in the optimization process.

Distributed State Estimation

The navigation error obtained as the objective of the contact plan optimization is based on covariance analysis simulation with simplified assumptions. To verify the positioning and timing accuracy of the optimized contact plan, we ran the proposed CI distributed state estimation algorithm (from Section 6) with the proposed two-way ranging and timing scheme (from Section 5). We used the contact plan with the lowest

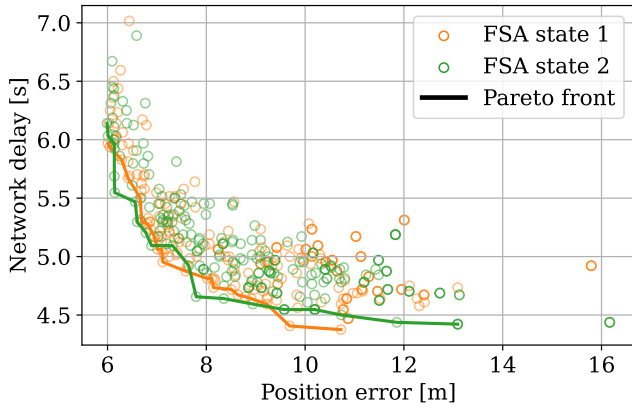


Figure 10: Network delay and navigation performance of the optimized contact plans for two FSA states, with the solutions in the Pareto front highlighted. Contact plans optimized for communications yield a 4.5-second BDT and over 10-meter positioning error, while those for navigation result in a 6-second BDT (25% worse) and a 6-meter error (40% better).

navigation error for FSA states 0 and 1.

The results are shown in Figure 11 and Figure 12. The position and clock bias estimation error converges within 2 m error, which is in the same order as the range and clock bias measurement errors we get in the two-way range method. The distributed CI algorithm generates consistent estimates with errors within the estimated covariance, as shown in Figure 12.

Table 3 shows the position, velocity, and clock bias estimation error for two different contact plans at both edges of the Pareto front. The plan optimized for navigation performance also shows lower state estimation errors than the plan optimized for BDT, which demonstrates the validity of the objective function based on covariance analysis to capture the preferable contact plan geometry.

8. CONCLUSION

Designing a contact plan for navigation satellite networks necessitates consideration of both inter-satellite ranging, which measures the distances between satellites for accurate positioning, and inter-satellite communication, enabling data exchange between satellites to form a global network. In this context, we target the complex problem of scheduling crosslinks within networks of small satellites in lunar orbit.

Similarly to previous work on Earth navigation satellite constellations, we frame the contact plan design problem as a constrained multi-objective optimization problem, considering both navigation and communications performance. However, we consider satellites with limited resources that need to rely on anchor nodes, i.e., a group of a few satellites with better navigation capabilities. Given the problem's complexity, we resort to a heuristic algorithm based on simulated annealing, a stochastic optimization method that gradually reduces the search space and decreases the likelihood of accepting suboptimal solutions. Simulation results show that our proposed method optimizes contact plans with reduced network delay and improved orbit determination accuracy, according to design criteria. Different solutions, however, present tradeoffs in terms of communications and navigation

performance, which need to be accounted for when designing and operating these constellations.

Finally, we verify the navigation performance of the optimized contact plans beyond the covariance analysis used in the optimization through a novel distributed orbit and clock bias estimation framework that combines an Extended Kalman Filter with a covariance intersection method. The filter processes two-way range and clock bias difference measurement considering light time delay, clock drifts, and large range-rates, being able to create consistent estimates given different contact plans.

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A: ALGORITHMS

Algorithm 2 shows the contact graph Dijkstra search algorithm that finds the route with best time delivery through a contact graph. Algorithm 3 shows the branch and exchange method, which is used to find the neighbor solution in simulated annealing.

Algorithm 2 Contact Graph Dijkstra Search. [27]

Input: Root contact C_{root} , destination D , contact plan U
Output: Route R_S^D from source S to destination D , and best delivery time BDT

```

1:  $R_S^D \leftarrow \{\}$ 
2:  $C_{fin} \leftarrow \{\}$  ▷ Final contact
3:  $BDT \leftarrow \infty$  ▷ Final arrival time
4:  $C_{curr} \leftarrow C_{root}$  ▷ Current contact is root
5: while True do ▷ Contact plan exploration loop
6:   ▷ Contact review procedure
7:    $C_{fin}, BDT \leftarrow \text{CRP}(U, C_{curr}, C_{fin}, BDT)$ 
8:   ▷ Contact selection procedure
9:    $C_{next} \leftarrow \text{CSP}(U, C_{curr}, BDT)$ 
10:  if  $C_{next}$  then
11:     $C_{curr} \leftarrow C_{next}$ 
12:  else
13:    break ▷ Review and selection done
14:  end if
15: end while
16: if  $C_{fin} \neq \{\}$  then
17:   $C = C_{fin}$ 
18:  while  $C \neq C_{S,S}^{0,\infty}$  do
19:     $R_S^D.\text{hops} \leftarrow \{C\}$ 
20:     $C = C_{pred}$  ▷ Previous contact
21:  end while
22:  Compute  $(R_S^D.\text{tx\_win}, R_S^D.\text{volume})$ 
23: end if
```

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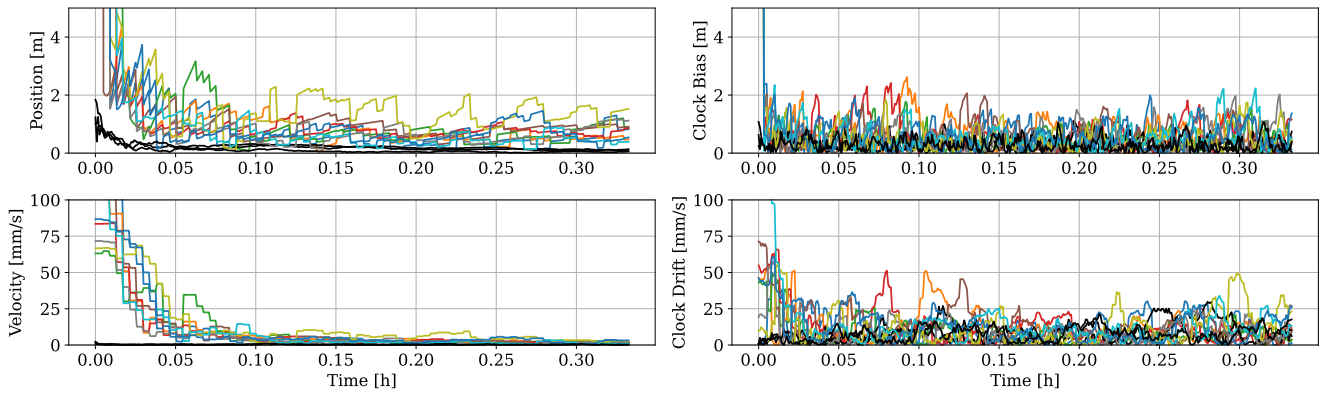


Figure 11: Position, velocity, clock bias, and clock drift estimation error obtained by the distributed state estimation algorithm. The contact plan with the lowest position error is used. Each line shows different satellites, and the simulation is for FSA states 0 and 1, which is 20 min in total.

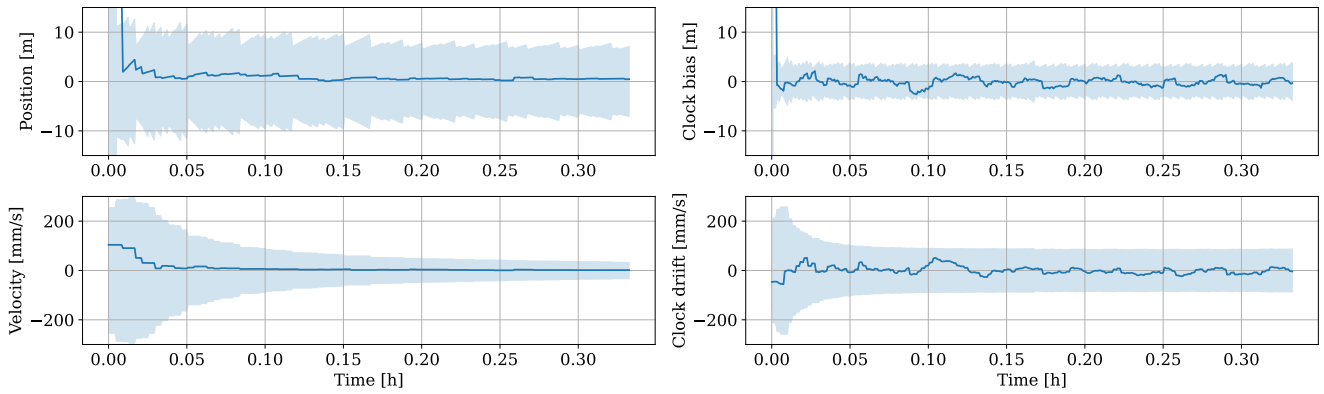


Figure 12: Position, velocity, clock bias, and clock drift estimation error (blue line) and the 3σ covariance estimate (blue shaded area) obtained by the distributed state estimation algorithm (Satellite 0). Consistent estimates are generated, with errors under the 3σ covariance estimate. The contact plan with the lowest position error is used, and the simulation is for FSA states 0 and 1, which is 20 min in total.

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Table 3: The obtained position, velocity, and clock bias estimation error with the distributed state estimation algorithm, with the two contact plans that are at the two edges of the Pareto front: The lowest navigation error plan and the lowest BDT plan. The final row shows the statistics among all satellites. The simulation is for FSA states 0 and 1. The lowest navigation error plan optimized based on the covariance analysis objective function also shows lower estimation error when use with the proposed CI distributed state estimation algorithm, showing the validity of the objective function to reflect the preference of contact geometry to improve positioning and timing. *Satellites 4, 6, and 11 are the anchor nodes, which gets positioning and timing aids from external sources.

Satellite	Lowest Navigation Error Plan			Lowest BDT Plan		
	Position Error [m]	Velocity Error [mm/s]	Clock Bias Error [m]	Position Error [m]	Velocity Error [mm/s]	Clock Bias Error [m]
0	3.27 ± 13.99	9.80 ± 16.34	1.23 ± 7.04	6.20 ± 13.67	20.16 ± 22.18	0.63 ± 1.27
1	4.23 ± 20.69	11.24 ± 27.09	0.68 ± 1.95	7.64 ± 14.94	17.48 ± 19.44	0.99 ± 4.89
2	3.05 ± 11.23	10.79 ± 15.40	0.88 ± 2.98	17.79 ± 15.01	34.06 ± 22.38	1.61 ± 9.37
3	3.24 ± 12.13	14.74 ± 21.90	0.74 ± 1.87	11.39 ± 15.50	67.24 ± 60.50	0.98 ± 1.87
4*	0.10 ± 0.20	0.38 ± 0.17	0.34 ± 0.25	0.16 ± 0.21	0.39 ± 0.15	0.36 ± 0.24
5	1.60 ± 3.88	9.37 ± 14.91	0.98 ± 4.39	5.81 ± 22.20	20.07 ± 28.22	1.15 ± 5.34
6*	0.09 ± 0.11	0.39 ± 0.23	0.34 ± 0.27	0.19 ± 0.16	0.35 ± 0.18	0.32 ± 0.24
7	2.72 ± 10.23	10.78 ± 25.57	0.66 ± 1.02	25.52 ± 22.29	29.75 ± 18.26	1.90 ± 12.50
8	3.65 ± 13.83	10.51 ± 10.39	0.78 ± 2.78	22.14 ± 19.22	30.19 ± 10.91	1.07 ± 4.40
9	3.85 ± 15.27	9.04 ± 9.88	0.78 ± 1.35	29.68 ± 28.37	35.41 ± 23.09	0.71 ± 0.60
10	2.38 ± 8.11	5.40 ± 5.33	1.55 ± 9.73	2.12 ± 6.49	11.14 ± 16.82	0.94 ± 4.31
11*	0.11 ± 0.11	0.30 ± 0.15	0.34 ± 0.23	0.21 ± 0.24	0.37 ± 0.16	0.38 ± 0.27
All	2.36 ± 1.46	7.73 ± 4.70	0.78 ± 0.35	10.74 ± 10.08	22.22 ± 18.37	0.92 ± 0.47

Algorithm 3 Branch and exchange method.

Input: Contact plan U , visibility matrix V

Output: New contact plan U_{new}

```

1:  $K_{perm} \leftarrow \text{randperm}(K)$ 
2: for  $k_{idx} \leftarrow 1$  to  $K$  do
3:    $k \leftarrow K_{perm}(k_{idx})$ 
4:    $S_{perm,1} \leftarrow \text{randperm}(S)$ 
5:    $S_{perm,2} \leftarrow \text{randperm}(S)$ 
6:   for  $i_{idx,1} \leftarrow 1$  to  $S$  do
7:      $i_1 \leftarrow S_{perm,1}(i_{idx,1})$ 
8:      $j_1 \leftarrow \text{argmax}(U(k, i_1, :))$ 
9:     if  $i_2 \leq i_1$  then
10:      continue ▷ symmetry
11:   end if
12:   for  $i_{idx,2} \leftarrow 1$  to  $S$  do
13:      $i_2 \leftarrow S_{perm,2}(i_{idx,2})$ 
14:      $j_2 \leftarrow \text{argmax}(U(k, i_2, :))$ 
15:     if  $V(i_1, j_2) = 0$  or  $V(i_2, j_1) = 0$  then
16:      continue ▷ visibility
17:   end if
18:   if  $\sum_{k'=1}^K U(k', i_1, j_2) > 0$  or  $\sum_{k'=1}^K U(k', i_2, j_1) > 0$  then
19:     continue ▷ ISL switching
20:   end if
21:    $U_{new} \leftarrow \text{Swap}(U(k, i_1, j_1), U(k, i_2, j_2))$ 
22:   break ▷ exchange completed
23: end for
24: end for
25: end for

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